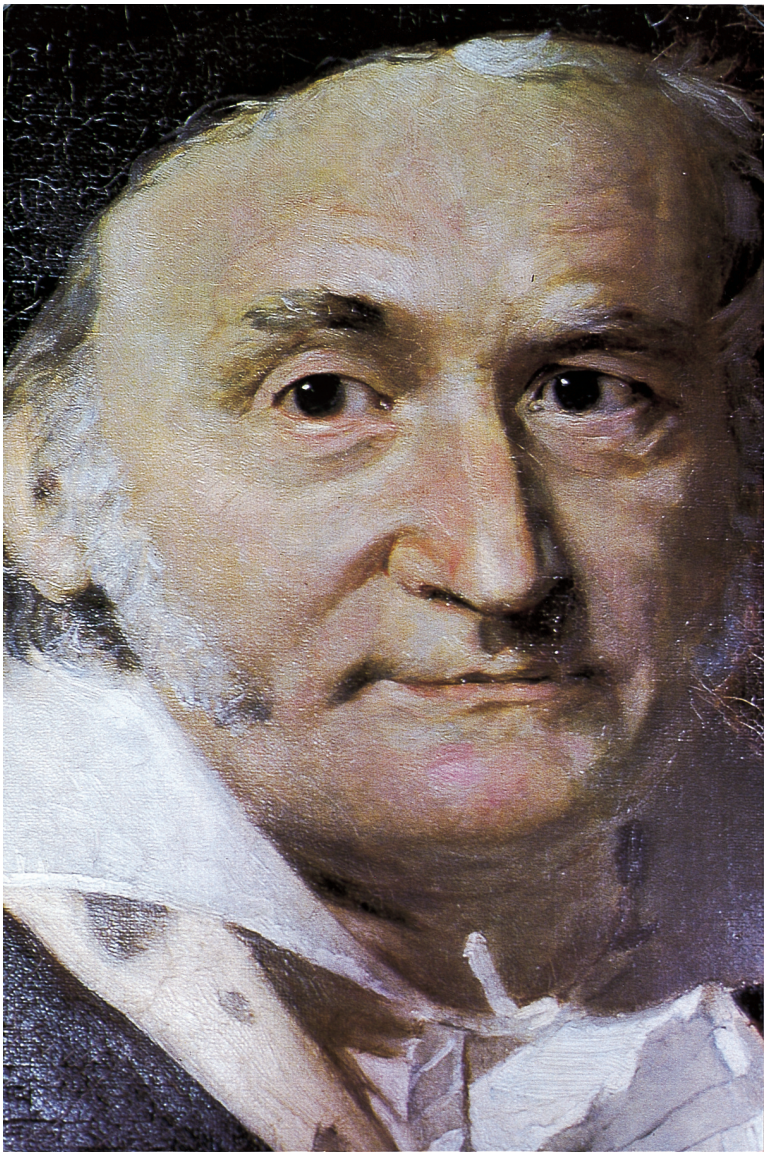
Part III

Gauß



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Carl Friedrich Gauß (1777–1855), the *Prince of Mathematicians*, was the latest, after Archimedes and Newton, in this trio of great men whose ideas shaped mathematics for centuries after their work (and two of whom figure prominently in this book).

Born on April 30, 1777, and registered as Johann Friderich Carl Gauß, he grew up in a poor bricklayer's family in Braunschweig. His father, an honest but tough and simple-minded person, did not succeed in keeping his son as uneducated as himself, mainly because of the efforts of Gauß' mother Dorothea and his uncle Friederich.

Gauß loved to tell the story of how—at ten years of age—one of the first flashes of his genius surprised his unsuspecting teacher Büttner. The class had been given the task to sum the numbers $1, \ldots, 100$. (What a useless task!) Gauß figured out the corresponding summation formula (see Section 23.1), wrote down the correct answer almost immediately, and waited while the other boys took the full hour to get their answers—all wrong. (Such stupidities have not vanished from German schools: the first author had a high-school geography teacher who would set similarly useless tasks in order to have some time for serious study—of the current *Playboy* issue.)

Luckily for Gauß, this teacher and others recognized his genius, and Duke Ferdinand of Braunschweig funded his high-school education and then university studies at Göttingen. This feudal generosity may have contributed to Gauß' absence of political involvement, although his were turbulent times: the French Revolution, Napoleonic Wars, and the revolutions of 1830 and 1848.

His student years 1795–1798 were incredibly productive. He found the ruler-and-compass construction of the regular 17–gon (and the determination, without proof, of which regular polygons can be constructed in this way), proved the law of quadratic reciprocity (where Euler's and Legendre's efforts had failed), discovered the method of least squares, classified binary quadratic forms, established the normal error distribution in probability theory (*Gauß' bell curve*), gave the first rigorous proof of the factorization of real polynomials into linear and quadratic factors (the *fundamental theorem of algebra*, his doctoral thesis in 1799), and much more.

He finished his masterpiece *Disquisitiones Arithmeticae* in 1798, at the age of 21 (Gauß 1801). Written in a terse style, according to Gauß' motto *pauca sed matura*[1], it was heavy going for the mathematicians of his age, but its wealth of ideas paved the way for new directions. Its publication was funded by the Duke; to limit the cost, Gauß had to leave out the last of eight planned parts, containing methods for the factorization of polynomials (see the Notes to Chapters

[1] few but mature

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14 and 15). This material was published posthumously from his handwritten notes (Gauß 1863b).

Gauß married Johanna Osthoff, younger by three years, in 1805. They had three children, but Johanna died after the last birth. Less than a year later, Gauß married her friend Minna Waldeck, eleven years younger, and they also had three children.

Gauß' influence permeates many parts of this book. His study of roots of unity and their subdivision according to subgroups of the relevant Galois group, the *Gauß periods*, can be seen as a precursor of the Fast Fourier Transform in Section 8.2. (These periods are also instrumental for modern fast algorithms for exponentiation in finite fields.) He proved the basic facts about factoring polynomials and the relation between factoring over $\mathbb{Z}$ and over $\mathbb{Q}$ (Section 6.2), found (but did not publish) the distinct-degree factorization method over finite fields (Section 14.2) and Hensel lifting (Section 15.4), guessed the prime number theorem (but did not prove it; see Notes 18.4), and studied hypergeometric series (Section 23.4). His *Gaussian elimination* is a staple of linear algebra (Sections 5.5 and 14.8).

Perhaps as important as his monumental contributions to so many fields is the fact that he championed the idea of mathematical rigor and watertight proof. This was often absent in 18th century mathematics, which lacked a precise understanding of things like limits and infinite sums. (Later, people such as Cauchy, Weierstraß, and Hilbert perfected Gauß' approach.)



According to himself, Gauß' work was only motivated by his inner urge for mathematical discoveries, and not his desire to publish or impress others. This manifested itself in markedly weak public relations. He did not educate a school of eager young disciples to spread his gospel, but he had a few brilliant students: Bernhard Riemann was his only pupil in the usual sense; Ferdinand Eisenstein and Richard Dedekind were his students in a wider sense. Many of Gauß' discoveries were not published during his lifetime: his insights on the arithmetic–geometric mean, elliptic functions and their double periodicity (with which Abel and Jacobi struggled later), the fundamental theorem on analytic functions (vanishing of closed curve integrals, rediscovered by Cauchy), quaternions (found by William Hamilton on 16 October 1843, when Gauß' notes had already slumbered in his drawers for thirty years), and his 1816 discovery of non-Euclidean geometry

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(given to the world by Nikolas Lobachevsky in 1829, and the son Johann Bolyai de Bolya of Gauß' friend Wolfgang Bolyai in 1832).

His appointment at the university of Göttingen in 1807 was as professor of astronomy. On 1 January 1801, Guiseppe Piazzi had discovered the asteroid Ceres—and it vanished in February. The astronomers could not find it again. Gauß used his newly devised computational methods in astronomy to calculate the orbit, and thanks to this, Ceres was rediscovered in December. This brought world-wide fame to him instantly. During his 48 years as professor, he gave 181 courses and seminars; of these, 128 were on astronomy, and only one on number theory.

One highly unusual aspect of Gauß' work is his uncanny mixture of theory and practice, with either profiting from the other. (Archimedes had a similar talent, while Newton's theoretical determination of improved ship hull cross-sections failed in practice.) This gave his scientific achievements a much wider audience than usual, and, after the low ebb of natural science (as opposed to literature, music, and philosophy) in Germany during the 18th century, he helped create an atmosphere in which bright young men were attracted to mathematics and science in the 19th century.

Gauß led, over many years, a geodetical survey of the Kingdom of Hannover. A private goal was to determine, in view of his discovery of non-Euclidean geometry, whether physical triangles really have an angle sum of 180 degrees—a question that astronomers still work on today with high-precision instruments. This work stimulated his research in differential geometry, leading to the important concept of Gaussian curvature and the Gauß–Bonnet theorem. He constructed, with Wilhelm Weber, an electric telegraph in 1833, with a 2-km-long wire, destroyed by lightning in 1845. He worked, also with Weber, on earth magnetism, and the unit of the magnetic force is called a *gauß*. At the Senate's request, he reorganized the University Widow's Fund, and on the way created the basis for modern life insurance calculations.

Gauß died in 1855, at age 77, and was buried in St. Albani's cemetery in Göttingen; today this is a pleasant park.

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Polynomial factorization is perhaps one of the
most striking successes of symbolic computation.

Zhuojun Liu and Paul S. Wang (1994)

La question de factorisation, que GAUSS considérait
avec raison comme *fondamentale*, est traitée dans notre ouvrage
avec une abondance de détails qu'on ne trouve pas
habituellement dans un livre d'étude, et certaines notions
y sont développées pour la première fois. [...]
Il ne nous appartient pas de nous prononcer sur la valeur scientifique
de notre exposé, mais nous avons la conviction
de n'avoir épargné, ni notre travail, ni notre temps,
pour élucider cette question importante.[1]

Maurice Kraïtchik (1926)

Le plus souvent, cependant, il sera aisé de trouver par le tâtonnement
une congruence irréductible d'un degré donné ν.[2]

Évariste Galois (1830)

In den meisten Wissenschaften pflegt eine Generation das
niederzureißen, was die andere gebaut, und was jene gesetzt,
hebt diese auf. In der Mathematik allein setzt jede Generation
ein neues Stockwerk auf den alten Unterbau.[3]

Hermann Hankel (1869)

Two bombs each. Every bomb had a 95 percent probability of hitting
[...] Even the paper probability was less than half a percent chance
of a double miss, but that times ten targets meant a five percent chance
that one missile would survive.

Tom Clancy (1994)

[1] The question of factorization, which GAUSS rightly considered as fundamental, is treated in our work with an abundance of details that one does not find usually in a textbook, and some notions are developed there for the first time. [...] It is not appropriate for us to make statements about the scientific value of our exposition, but we have the conviction that we have spared neither effort nor time in order to elucidate this important question.

[2] Most often, however, it will be easy to find by trial an irreducible polynomial [modulo a prime number] of a given degree ν.

[3] In most sciences one generation tears down what another has built and what one has established another undoes. In mathematics alone each generation builds a new storey on top of the old structure.

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